nanoker

CERAMIC TECHNICAL DATA SHEET

KA-EDM

Electroconductive Alumina Composite

KEY PROPERTIES

- Very high hardness
- Excellent strength / thoughness balance
- High thermal conductivity
- Ra < 0.1 μm after mirror polishing
- Machinable by EDM (electro-discharge machining)

APPLICATIONS

- Cutting tools
- Wear resistant tools for stamping
- Cold drawing dies
- Machinery parts
- Ballistic protection
- Chemical industry

MAIN PROPERTIES

Property	Value	Unit
Density	4.2	g/cm³
Flexural Strength	850	MPa
Hardness	22.5	GPa
Fracture Toughness	8.0	MPa·Vm
Young Modulus	370	GPa
Thermal Conductivity (100°C)	35	W/m·K
CTE (20–1000ºC)	8.0	$x10^{-6}$ K^{-1}

** All properties measured at 20ºC unless otherwise stated.*
*** Measured in a wall thickness of 2 mm*

OPERATING CONDITIONS

Max working temp (air)	1000	ºC
Recomended temp	<950	ºC
Continuous temp	900	ºC

ELECTRICAL BEHAVIOR

Resistivity	$3.1 \cdot 10^{-5}$	Ω·cm
Conductivity	High	

MATERIAL INFORMATION

Material type	Electroconductive ceramic
Machinability	EDM machinable
Structure	Composite ceramic
Key facture	Conductive

Local overheating may occur at electrical contact points due to current concentration and geometry effects.

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